PERARASAN D

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EDUCATION

SRM Institute of Science and Technology **2024 - 2026**
M.Sc Applied Data Science **GPA:** 9.15/10.
Agurchand Manmull Jain College **2021 - 2024**
Bachelor of Computer Applications **GPA:** 7.74/10.

TECHNICAL SKILLS

Programming & Tools: Python, C++, SQL, Git, GitHub, Streamlit, Flask, FastAPI, Tableau
Machine Learning & AI: NumPy, Pandas, SciPy, Statsmodels, Matplotlib, Seaborn, Deep Learning (PyTorch, TensorFlow, Keras), Computer Vision (OpenCV, YOLO), NLP (NLTK, spaCy), GANs
Generative AI: LangChain, LangGraph, FAISS, Pinecone, Google ADK, MCP (Model Context Protocol)
Cloud & DevOps: AWS (S3, EC2, ECS, Lambda, SageMaker), Docker, CI/CD (GitHub Actions), MLflow

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WORK EXPERIENCE

AI Intern
FEB 2026 - MAY 2026
Meraki Alai Labs, Chennai

- Built a LangGraph-orchestrated workflow automation system in Python that fully automated the daily report downloading process across 3 modules and 5 dealerships (85 reports/day), reducing operational time from 5–7 hours to 25–30 minutes per day — a 100%+ reduction in manual effort — eliminating errors in time-sensitive reporting pipelines.
- Developed and deployed a RAG application running entirely on-device on a mobile phone using Gemma 4 LLM, enabling fully offline, private document question-answering without any cloud dependency.
- Engineered production-grade object detection and classification models spanning the full ML lifecycle — data collection, labeling, training, and evaluation across audio classification and multi-class visual scene understanding for real-world robotics and automation systems.
- Conducted R&D across computer vision and NLP domains, translating experimental findings into scalable, production-ready solutions that directly improved client product performance and reliability.

PROJECTS

Bone Vision

- Built an end-to-end multimodal deep learning system on hand X-rays that jointly predicts Age (MAE 0.04), Gender (96.72% accuracy), and Bone Health Index (BHI) in under 5 seconds — replacing a manual workflow that takes radiologists 7–10 minutes per scan.
- Engineered a multi-task EfficientNet-B3 with attention, achieving 96.72% gender accuracy (AUC 0.9938) and age MAE of 0.0407, alongside a custom YOLO metacarpal detector reaching mAP@50 99.5%, mAP@50–95 94.86%, recall 1.0, and precision ~0.998 for clinical-grade BHI computation.
- Shipped as a full-stack app with PyTorch models behind a Flask API and a drag-and-drop web UI (HTML/CSS/JS) supporting both DICOM and standard image formats

Autonomous Data Analyst Agent

- Built a self-correcting LangGraph agent that writes and executes its own SQL/Python to answer natural-language questions on CSV/Excel data.
- Multi-tool orchestration across DuckDB, sandboxed Python, and Plotly; Streamlit UI with Redis-backed session memory.

Document Q&A System with RAG

- Built production-ready RAG system achieving 98% source attribution accuracy with 20–30s CPU response times using FAISS vector search, LangChain chunking, and TinyLlama inference with CPU-optimized pipeline. •
- Full-stack: FastAPI + Streamlit, 5+ document formats, under 10s upload time, near-zero cost.

Customer Churn Prediction

- Built an end-to-end churn prediction system achieving 82.4% ROC-AUC and 82% recall on 7K+ telco records; performed feature engineering on 30 variables and applied class balancing to handle 73–27% imbalanced dataset.
- Engineered production-grade ML pipeline with FastAPI, Streamlit dashboard, Docker, and GitHub Actions CI/CD; implemented MLflow for experiment tracking and automated deployment across 6 models.

CERTIFICATES

- **Complete Machine Learning and Data Science** From GeeksforGeeks
- **Mathematics for Machine Learning and Data Science Specialization** from DeepLearning.AI
- **IBM Rag and Agentic AI** from IBM